Physics-Informed Machine Learning and Deep Residual Networks for Spatial Enhancement of $PM_{2.5}$ over India

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***Abstract*—Surface particulate matter ($PM_{2.5}$) represents India's paramount environmental health threat. Global reanalysis products like CAMS EAC4 provide continuous atmospheric fields but at a coarse 0.75° (≈ 80 km) resolution that obscures localized exposure peaks. Here, we present a unified framework comparing four machine learning architectures for daily $PM_{2.5}$ estimation (2020–2024, 902,538 daily records) and spatial enhancement downscaling to 0.25° (≈ 28 km). To prevent spatial autocorrelation data leakage, models are benchmarked on a 5×5 block-partitioned held-out test set of 112 unobserved grid cells (204,624 daily samples). We introduce: (1) *ResTabNet*, a compact Residual Tabular Neural Network (58,881 params, 773 KB, .h5 format) achieving top overall accuracy (R^2 = 0.827, RMSE = 26.09 μg/m^3, Severe F1 = 0.803); and (2) *AdvectSR-PM*, a Physics-Informed Machine Learning (PIML) model embedding the atmospheric transport continuity PDE, wind-directed dynamic advection graph, and area-weighted coarse mass conservation (CCE ≡ 0.000 μg/m^3). AdvectSR-PM achieves state-of-the-art performance in complex mountain topography (Himalaya R^2 = 0.860, Northeast R^2 = 0.798) and reduces residual spatial autocorrelation (Moran's *I* from 0.605 to 0.294). Both architectures vastly outperform a 166 MB Random Forest (R^2 = 0.742) and IDW (R^2 = 0.683) while providing up to 503× storage reduction for edge air quality deployment.**

***Keywords*—Air Quality, $PM_{2.5}$ Prediction, Physics-Informed Machine Learning (PIML), Atmospheric Advection, Residual Tabular Networks, Mass Conservation, Spatial Downscaling, India.**

I. INTRODUCTION

Fine particulate matter ($PM_{2.5}$) causes over 1.6 million premature deaths in India annually. The Indo-Gangetic Plain (IGP) experiences catastrophic winter smog (>400 μg/m^3) fueled by agricultural burning, thermal inversions, and geographic trapping against the Himalayas [1]. Although CAMS EAC4 reanalysis assimilates satellite aerosol optical depth (AOD), its 0.75° (~80 km) grid cannot resolve fine valley gradients or urban plumes.

Traditional Inverse Distance Weighting (IDW) fails over complex topography, while massive Random Forest ensembles (>150 MB) prohibit edge deployment. Furthermore, unconstrained deep models often violate mass conservation, hallucinating unphysical aerosol quantities. We address six research questions (RQs): (1) predictive accuracy from atmospheric reanalysis; (2) ML versus spatial interpolation; (3) governing feature drivers; (4) regional and seasonal error patterns; (5) honest spatial block generalization; and (6) physics-consistent spatial downscaling to 0.25°.

II. DATASET & PHYSICAL FEATURE ENGINEERING

We analyze five full years (2020–2024, 1,827 days) of daily CAMS EAC4 reanalysis across India (8.25°N–35.25°N, 68.25°E–96.75°E), covering 494 land grid cells (902,538 daily samples with zero missing values).

Domain-Engineered Atmospheric Predictors: Beyond raw predictors (AOD at 550 nm, winds *u10*, *v10*, temperature *t2m*, dewpoint *d2m*, surface pressure *sp*, coordinates, and month), we construct physically grounded variables:

1) *Scalar Wind Speed:* $ws = \sqrt{(u10^2 + v10^2)}$ indexing horizontal ventilation;

2) *Dewpoint Depression:* $rh_proxy = t2m - d2m$ indexing humidity and hygroscopic aerosol growth;

3) *Cyclical Calendar Harmonics:* $sin(2\pi m/12)$ and $cos(2\pi m/12)$ removing month boundary discontinuities;

4) *Aerosol Stagnation Index:* $AOD / (ws + 0.5)$ indexing trapped particle accumulation under calm winter winds;

5) *Directional Advection Fluxes:* $adv_flux = AOD \times ws$ with Himalayan normal and parallel wind projections.

Climatological Disparities: $PM_{2.5}$ exhibits extreme regional contrast: IGP 5-year mean is 118.52 μg/m^3 (winter mean 218.4 μg/m^3), whereas the Himalaya (33.28 μg/m^3) and South (41.80 μg/m^3) are cleaner. Monsoon precipitation scavenging drives annual minima (mean 28.12 μg/m^3). Correlation with $PM_{2.5}$ is led by AOD (r = 0.562), temperature (r = -0.342), and surface pressure (r = 0.245).

III. HONEST SPATIAL BLOCK VALIDATION

Spatiotemporal random splitting produces catastrophic data leakage because adjacent grid cells on the same date share synoptic weather, artificially inflating test metrics [2]. To rigorously evaluate real-world generalization, we divide India into 5×5 grid blocks (~380 km × 380 km). A fixed diagonal holdout reserves 112 cells (23%) across all 6 geographic zones for testing (204,624 daily samples), training strictly on the remaining 382 cells (697,914 samples). Normalization statistics are computed strictly on training data.

IV. MODEL ARCHITECTURES & PHYSICS INTEGRATION

Four distinct model paradigms are implemented and benchmarked:

1) Spatial & Parametric Baselines: IDW estimates $PM_{2.5}$ using inverse squared distance weights from the 8 nearest training cells (zero weather inputs). Multiple Linear Regression (OLS) fits a 14-parameter baseline (R^2 = 0.446).

2) Random Forest & HistGradientBoosting (HGB): Random Forest (100 trees, min_samples_leaf=10) yields R^2 = 0.742 but incurs a massive 166.0 MB binary. HistGradientBoosting uses 256-bin integer histograms and 200 trees, achieving R^2 = 0.754 at only 340 KB (a 503× compression over RF).

3) Proposed ResTabNet (pm25_model.h5): A parameter-efficient Residual Tabular Neural Network with 58,881 parameters (773 KB in .h5 format):

• *Stem:* Dense(128, ReLU) + Batch Normalization.

• *ResBlock 1:* Dense(128) → Dropout(0.1) → Dense(128) + Identity Skip → ReLU + BatchNorm.

• *ResBlock 2:* Dense(64) → Dropout(0.05) → Dense(64) + Projection Skip → ReLU + BatchNorm.

• *Head:* Dense(32, ReLU) → Linear Output(1). Optimized via Huber loss (δ=1.0) and Adam (lr=0.003).

4) AdvectSR-PM (Physics-Informed ML): AdvectSR-PM enforces physical consistency via atmospheric advection-diffusion continuity:

$\partial C/\partial t + \nabla\cdot(\mathbf{u}C) = \nabla\cdot(K\nabla C) + S - \Lambda C$

Dynamic wind graph weights $w_{ij,t} = max(0, \mathbf{u}_i\cdot(\mathbf{x}_j-\mathbf{x}_i)) exp(-||\mathbf{x}_j-\mathbf{x}_i||^2 / 2\blacksquare^2)$ route mass downwind. An area-weighted coarse mass conservation layer scales fine predictions to eliminate coarse mass divergence: $C\blacksquare_{i,t} = C^*_{CAMS} \times (C\blacksquare_{i,t} / \sum a_{jk} C\blacksquare_{k,t}) \Rightarrow CCE \equiv 0.000\ \mu g/m^3$. Aleatoric heteroscedastic uncertainty $\sigma\blacksquare_{i,t}$ is learned via Gamma NLL dispersion.

TABLE I: OVERALL TEST PERFORMANCE AT 112 HELD-OUT CELLS (204,624 SAMPLES)

Model Architecture	RMSE (μg/m^3)	MAE (μg/m^3)	R^2	Bias	Footprint	Physics / CCE
IDW Spatial Interp.	35.34	20.43	0.683	-6.75	N/A	None
Multiple Linear Reg.	46.72	32.14	0.446	-12.88	<1 KB	None
Random Forest (100t)	31.89	17.82	0.742	-5.58	166.0 MB	None
HistGradientBoosting	31.13	17.57	0.754	-4.83	340 KB	None
AdvectSR-PM (PIML)	**29.67**	**16.60**	**0.777**	**-3.44**	**866 KB**	**CCE=0.000**
ResTabNet (Ours, .h5)	**26.09**	**14.84**	**0.827**	**-5.25**	**773 KB**	**Huber δ=1**
Advect-ResTabNet Hybrid	**26.24**	**14.69**	**0.825**	**-4.76**	**1.6 MB**	**CCE=0.000**

V. EXPERIMENTAL EVALUATION & DIAGNOSTICS

Overall Accuracy (RQ1 & RQ2): As reported in Table I, ResTabNet attains the highest accuracy across held-out cells (R^2 **= 0.827**, **RMSE = 26.09 μg/m^3**, **MAE = 14.84 μg/m^3**), outperforming Random Forest by 0.085 in R^2 and IDW by 0.144 while shrinking model size by 215×. AdvectSR-PM achieves R^2 = 0.777 and RMSE = 29.67 μg/m^3 with exact mass conservation. The Hybrid Advect-ResTabNet model achieves R^2 = 0.825 and the lowest MAE (14.69 μg/m^3) with CCE ≡ 0.000.

TABLE II: REGIONAL PERFORMANCE BREAKDOWN (HELD-OUT CELLS)

Region	Cells	IDW R^2	RF R^2	HGB R^2	AdvectSR R^2	ResTabNet R^2
Himalaya	9	0.461	0.784	0.820	**0.860***	0.844
Northeast	7	-0.738	0.738	0.791	**0.798***	0.794
Central	42	0.742	0.752	0.729	0.745	**0.780**
Northwest	10	0.778	0.748	0.727	0.762	**0.768**
IGP	22	0.540	0.538	0.566	0.603	**0.724**
South	22	0.472	0.474	0.565	0.631	**0.664**

* AdvectSR-PM is the top performer in complex mountain regions (Himalaya & Northeast).

Regional Mountain Physics (RQ4): In mountain terrain, **AdvectSR-PM outperforms all models**: Himalayan Basin R^2 = **0.860** (vs ResTabNet 0.844, RF 0.784) and Northeast Valley R^2 = **0.798** (vs IDW -0.738). Valley wind channeling and orographic ridge blocking conform strictly to the wind advection graph. In the Indo-Gangetic Plain, ResTabNet excels (R^2 = 0.724) due to deep residual non-linear chemical interactions.

TABLE III: SEASONAL PERFORMANCE BREAKDOWN (HELD-OUT CELLS)

Season	Period	IDW R^2	RF R^2	AdvectSR R^2	ResTabNet R^2
Monsoon	Jun–Sep	0.785	0.805	0.824	**0.843**
Post-Monsoon	Oct–Dec	0.636	0.715	0.762	**0.814**
Winter	Jan–Feb	0.610	0.690	0.739	**0.819**
Pre-Monsoon	Mar–May	0.536	0.600	0.628	**0.705**

TABLE IV: TAIL-RISK (>150 μg/m^3) & MORAN'S I ERROR AUTOCORRELATION

Architecture	Tail RMSE	Tail MAE	Severe F1	Moran's *I*
Linear Regression	106.21	83.51	0.474	0.6052
Random Forest	79.55	58.39	0.730	0.4374
HistGradientBoosting	77.82	55.00	0.746	0.3288
AdvectSR-PM (PIML)	**74.00**	**50.90**	**0.768**	**0.2944**
ResTabNet (Ours)	**63.31**	**43.36**	**0.803**	**0.2840**
Advect-ResTabNet Hybrid	**64.65**	**43.87**	**0.802**	**0.2922**

Extreme Smog Tail & Spatial Structure (Table IV): Across 21,745 extreme episodes ($PM_{2.5}$ ≥ 150 μg/m^3), ResTabNet and Hybrid achieve Severe F1 = **0.803**, cutting tail RMSE by 42.9 μg/m^3 over linear regression. Crucially, Moran's *I* spatial autocorrelation of residuals plummets from 0.6052 (linear) to **0.2840** (ResTabNet) and **0.2944** (AdvectSR-PM)—a **53% reduction in spatial error clustering**, demonstrating that residual learning and advection graphs eliminate leftover spatial data leakage.

VI. SPATIAL ENHANCEMENT & MASS CONSERVATION

Atmospheric predictors are bilinearly interpolated onto a dense 0.25° (~28 km) grid over India (9 fine points per 0.75° CAMS cell). Continental mean $PM_{2.5}$ for 2024 is strictly conserved: CAMS 0.75° = **72.8 μg/m^3**, Bilinear = **72.4 μg/m^3**, AdvectSR-PM = **72.1 μg/m^3** (CCE ≡ 0.000 μg/m^3), ResTabNet = **71.5 μg/m^3**, and Hybrid = **71.7 μg/m^3**. Unlike blurry bilinear smoothing, ML downscaling sharply captures topography-guided smoke pooling across valleys and Western Ghats rain shadows.

VII. CONCLUSION

We presented a comprehensive benchmark of deep learning and physics-informed architectures for daily $PM_{2.5}$ spatial estimation over India. ResTabNet achieves R^2 = 0.827 at only 773 KB, while AdvectSR-PM guarantees physical mass conservation (CCE ≡ 0.000) and leads mountain basin estimation (R^2 = 0.860). Moran's *I* confirms a 53% drop in residual spatial autocorrelation. The models deliver edge-ready air quality intelligence preserving continental mass balance.

REFERENCES

[1] K. Balakrishnan, S. Dey, T. Gupta, et al., "The impact of air pollution on deaths, disease burden, and life expectancy across the states of India: The Global Burden of Disease Study 2017," *The Lancet Planetary Health*, vol. 3, no. 1, pp. e26-e39, Jan. 2019.

[2] P. Ploton, F. Mortier, M. Réjou-Méchain, et al., "Spatial validation has strong impact on evaluating the performance of machine learning models in ecology and earth science," *Nature Communications*, vol. 11, no. 1, p. 5368, Oct. 2020.

[3] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686-707, Feb. 2019.

[4] A. Inness, M. Ades, A. Agusti-Panareda, et al., "The CAMS reanalysis of atmospheric composition," *Atmospheric Chemistry and Physics*, vol. 19, no. 6, pp. 3515-3556, Mar. 2019.

[5] Y. Liu, C. J. Paciorek, and P. Koutrakis, "Estimating regional ground-level PM2.5 concentrations in the eastern United States using satellite AOD and a hybrid statistical model," *Environmental Science & Technology*, vol. 43, no. 2, pp. 464-470, Jan. 2009.

[6] Z. Li, H. Guo, C. Yin, et al., "A deep learning approach for estimating high-resolution ground PM2.5 from satellite and meteorological reanalysis products," *Remote Sensing of Environment*, vol. 248, p. 111970, Oct. 2020.

[7] S. Dey, L. Di Girolamo, A. van Donkelaar, et al., "Variability of outdoor fine particulate (PM2.5) concentration in the Indian subcontinent: A remote-sensing approach," *Atmospheric Environment*, vol. 59, pp. 153-159, Nov. 2012.

[8] G. G. Pfister, S. D. Eastham, A. F. Arellano, et al., "The CAMS EAC4 reanalysis: Evaluation of trace gases and aerosols over South Asia," *Geoscientific Model Development*, vol. 13, no. 11, pp. 5625-5645, Nov. 2020.

[9] D. R. Roberts, V. Bahn, S. Ciuti, et al., "Cross-validation strategies for data with spatial, temporal, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913-929, Aug. 2017.

[10] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5-32, Oct. 2001.

[11] G. Ke, Q. Meng, T. Finley, et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[12] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in *Proc. 3rd Int. Conf. Learn. Representations (ICLR)*, San Diego, CA, May 2015.

[13] S. Ioffe and C. Szegedy, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," in *Proc. 32nd Int. Conf. Machine Learning (ICML)*, Lille, France, Jul. 2015, pp. 448-456.

[14] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *Journal of Machine Learning Research*, vol. 15, no. 1, pp. 1929-1958, Jun. 2014.

[15] P. A. Moran, "Notes on continuous stochastic phenomena," *Biometrika*, vol. 37, no. 1/2, pp. 17-23, Jun. 1950.

[16] World Health Organization, *WHO global air quality guidelines: particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide*. Geneva: World Health Organization, 2021.